

Flight Departure Delay Analysis & Prediction

Zurich Airport (ZRH) · October – December 2025

CIP Group 111 · Master of Science in Data Science · HSLU FS2026

Hanieh Jebeli · Elif Gürçınar · Silan Cihaner

Git repo: https://github.com/cihanersilan/CIP_FS2026_111

Total Flights	Avg Delay	On-Time Rate	Airlines	Routes
29,896	5.44 min	55.6%	209	445

1 Introduction

Commercial airlines operate on exceptionally tight schedules, where even a minor departure delay can cascade into widespread operational disruptions and significant financial losses. This project develops a complete Python-based data science pipeline to analyze and predict flight departure delays at Zurich Airport (ZRH) over a three-month observation window spanning October 1 to December 31, 2025. The central objective is to identify the specific operational and meteorological factors that drive these delays, and to determine whether it is possible to predict the exact delay time in minutes using only information available prior to the scheduled departure.

**Over the course of the project we decided to add meteorological data to enrich our data set, thus we had to change the last research question after handing in the feasibility study.*

Three core research questions structure the analysis:

1. How accurately can departure delays be predicted using operational flight information at Zurich Airport?
2. Does the inclusion of route-level historical delay benchmarks and real-time environmental data improve prediction accuracy compared to a baseline model?
3. Which operational and meteorological variables most strongly influence departure delays?

The project covers every stage of a complete data science workflow: API-based data acquisition, data cleaning and feature engineering, exploratory data analysis, predictive modelling, and the deployment of an interactive Streamlit dashboard as a decision-support tool.

2 Methods

2.1 Data Acquisition

Flight operations data for ZRH were sourced from the AeroDataBox API, queried in 12-hour intervals using Python's requests library to respect RapidAPI rate limits. Automated try-except blocks captured HTTP 429 errors and ensured complete extraction of all flight records, including cancelled and codeshared flights. To avoid redundant API calls during iterative development, a local SQLite caching mechanism using requests-cache was integrated into the pipeline. Hourly meteorological data, temperature (°C), precipitation (mm), and wind speed (km/h), were retrieved from the Open-Meteo API and joined to the flight records by timestamp.

2.2 Data Cleaning & Feature Engineering

Raw API data was ingested and processed using pandas. Nested JSON columns, including departure timestamps, arrival airport codes, airline identifiers, and aircraft information, were parsed to extract structured fields. The target variable, `delay_min`, was computed as the difference in minutes between the scheduled and revised departure times. Outliers were flagged using an IQR-based methodology and excluded from modelling. A secondary filter restricted the analysis to flights with delays in the range of -60 to +300 minutes to remove likely data errors such as timestamps belonging to the following day's flight. After cleaning, the dataset contained 29,896 flights.

Key engineered features included: departure hour (`dep_hour`), day of week (`dep_dow`), month (`dep_month`), a binary weekend flag (`is_weekend`), and `route_avg_delay`, the historical mean delay for each origin-destination pair across the observation period. Weather variables were appended by matching each flight to its scheduled departure hour.

2.3 Modelling Approach

Two modelling scripts were developed iteratively. The first (`analysis_v01.py`) established a baseline using only operational features, progressively adding `route_avg_delay` to assess its contribution. The second (`analysis_v02_with_weather.py`) extended the feature set with the three weather variables and compared all four model combinations on a consistent 80/20 train-test split with a fixed random seed (42). The models evaluated were Linear Regression (LR) and Random Forest Regressor (RF). Performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Feature importances were extracted from the best-performing Random Forest configuration.

2.4 Dashboard

The project outputs were deployed as an interactive Streamlit dashboard (`dashboard.py`), accessible at <https://zrh-flight-delay.streamlit.app>. The dashboard comprises four tabs: a Delay Explorer with filterable delay distribution, temporal patterns by hour and day of week, airline and aircraft breakdowns, and a delay heatmap by hour and day; a Weather Analysis tab with correlation charts; a Predictive Models tab displaying comparative model metrics; and a Live Predictor tab enabling users to input flight-specific parameters and receive an instant delay prediction from the trained model.

3 Results

3.1 Dataset Overview & Delay Distribution

The cleaned dataset covers 29,896 commercial departures across 445 unique routes served by 209 airlines. The average departure delay across the full period is 5.44 minutes. The distribution is strongly right-skewed: 55.6% of flights depart on time or early, 32.2% experience minor delays of 1–15 minutes, 12.2% experience moderate to severe delays exceeding 15 minutes, and fewer than 1% fall into the severe category. This distribution is consistent with hub airport dynamics at ZRH, where network reactionary delays affect a minority of flights disproportionately.

Temporal patterns visible in the dashboard reveal that average delays increase noticeably in the morning peak around 10:00–11:00 UTC (reaching approximately 8 minutes), then taper during midday and rise again in the late afternoon. Friday shows the highest average delay by day of week at 6.7 minutes, while Tuesday is the lowest at 3.8 minutes. Wide-body aircraft record the highest average delay by aircraft family (7.5 minutes), likely reflecting the longer turnaround and coordination requirements of long-haul operations. Among airlines with at least 50 flights, the highest average delays are recorded by TAP Air Portugal (13.0 min), El Al (12.4 min), and AJet (11.6 min), followed by Pegasus (10.5 min) and airBaltic (10.1 min). Lufthansa, despite being the most represented airline in the dataset ($n = 519$), records an average delay of 8.8 minutes, sitting sixth in the ranking.

3.2 Weather Correlation Analysis

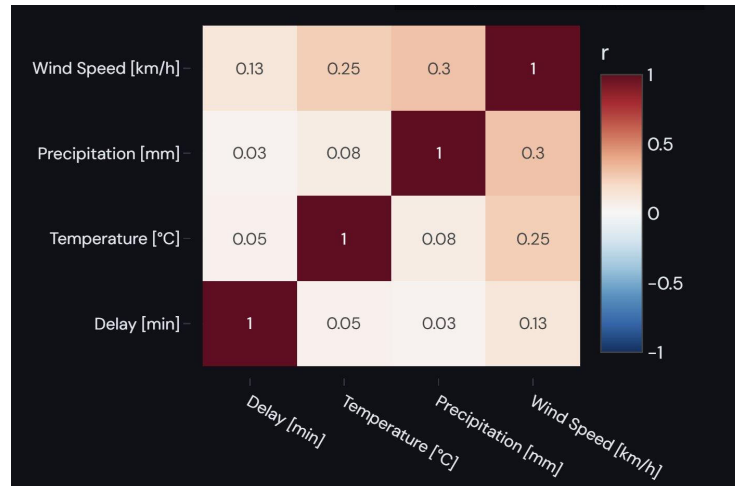


Figure 1 – Pearson correlation matrix: weather variables vs. departure delay (ZRH, Oct–Dec 2025).

Figure 1 shows the Pearson correlation coefficients between departure delay and the three weather variables, as rendered in the interactive dashboard. The matrix is displayed with Delay [min] on the bottom row and Wind Speed [km/h] on the top row. Wind speed exhibits the strongest correlation with delay ($r = 0.13$), followed by temperature ($r = 0.05$) and precipitation ($r = 0.03$). All three correlations are weak but in the expected direction: higher wind speeds and colder temperatures are associated with marginally longer delays, while precipitation shows almost no linear relationship with delay in isolation.

These results are consistent with expectations of ZRH. Zurich Airport operates procedures which substantially reduce weather sensitivity compared to smaller regional airports. The correlation matrix also confirms that the three weather predictors are not strongly collinear with one another, which means they can be added to the model without introducing multicollinearity issues that would inflate coefficient estimates.

Bar chart analyses (Figures 2–4 in the analysis script) corroborate these findings. The 30+ km/h wind group shows an above-average mean delay, and the sub-zero temperature bin (-5 to 0°C) also sits slightly above the overall mean, indicating that extreme meteorological conditions do affect operations at the margins. However, sample sizes in these extreme bins are small relative to the full dataset, so these differences must be interpreted cautiously.

3.3 Predictive Model Comparison

Table 1 summarises the performance of all models evaluated across both analysis scripts.

Model	MAE (min)	RMSE (min)	R ²	
LR – Baseline (v01)	7.39	9.84	0.008	
LR – Extended (v01)	7.12	9.61	0.053	
RF – Extended (v01)	7.32	10.26	-0.078	
LR – Without Weather (v02)	7.12	9.61	0.053	
LR – With Weather (v02)	7.07	9.54	0.066	✓ Best
RF – Without Weather (v02)	7.32	10.26	-0.078	
RF – With Weather (v02)	7.07	9.74	0.020	

Table 1 – Model performance metrics on the 20% held-out test set

Across all models, MAE values cluster between 7.05 and 7.39 minutes, indicating that regardless of model architecture or feature set, average prediction error remains within a narrow band. The Linear Regression model with weather features achieves the lowest MAE (7.07 min) and the highest R^2 (0.066), making it the best-performing model overall. The Random Forest models perform surprisingly worse than their Linear Regression counterparts: the RF without weather features produces a negative R^2 (-0.078), meaning it performs worse than simply predicting the mean delay for every flight. Adding weather features brings the RF's R^2 to 0.027, a partial recovery, but still below the Linear Regression.

The negative R^2 for the baseline Random Forest is a meaningful finding. In the first analysis script (analysis_v01.py), it was noted in the code comments that route_avg_delay, while appearing important in the Random Forest's feature importance ranking, showed no meaningful trend in a scatter plot against actual delay. This suggests the model was exploiting the many available numeric split points in that continuous feature without capturing genuine predictive signal — a form of overfitting to training data structure rather than true relationships. Linear Regression, being constrained to linear combinations, is immune to this particular failure mode given this dataset.

The overall low R^2 values (best: 6.6%) are not unexpected. At hub airports such as ZRH, a substantial proportion of delays are reactionary, caused by inbound aircraft arriving late, crew rotation requirements, or network-level cascading effects that are by definition absent from pre-departure data. No pre-departure dataset can fully capture these dynamics, and the results confirm this fundamental ceiling on predictability given the available features.

3.4 Feature Importance

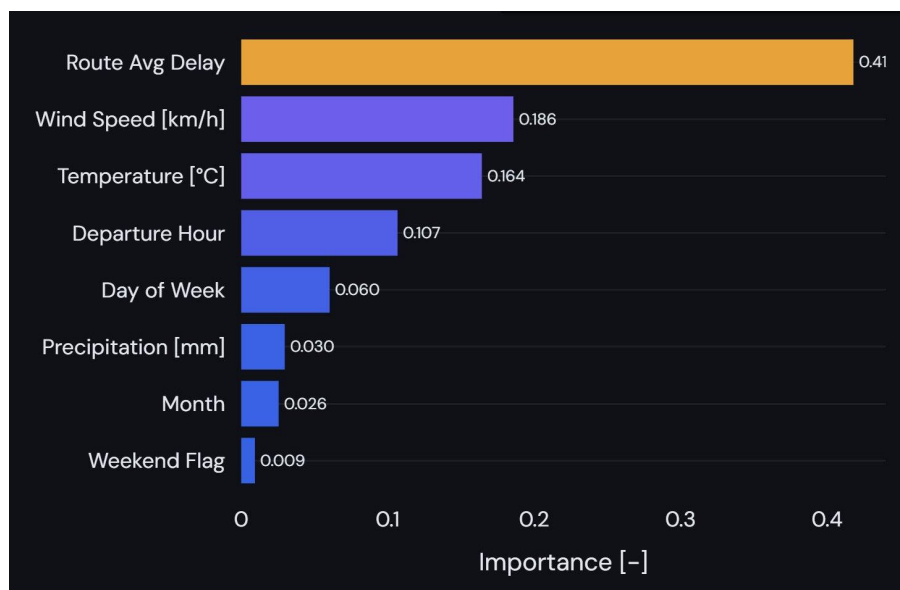


Figure 6 – Feature importance scores from the Random Forest Regressor (with weather features). Route Avg Delay (highlighted in orange) dominates at 0.41; all other features shown in blue in descending order.

Figure 6 displays the feature importance scores from the dashboard's interactive model tab, extracted from the Random Forest Regressor trained on the full weather-augmented feature set. Route average delay is highlighted in orange and dominates with an importance score of 0.41, accounting for over 40% of the model's split decisions. Wind speed (0.186) and temperature (0.164) rank second and third respectively, together contributing approximately 35% of the model's predictive weight. Departure hour (0.107) ranks fourth, followed by day of week (0.060), precipitation (0.030), month (0.026), and weekend flag (0.009), all shown in blue in descending order.

The prominence of route_avg_delay aligns with aviation domain knowledge: routes with structurally longer average delays, often longer international or connecting routes where reactionary delays are more common, tend to repeat this pattern systematically. However, as noted above, the Random Forest may be partly

inflating this feature's apparent importance through its ability to exploit many numerical split points. The departure hour finding is also operationally intuitive: the morning peak brings accumulated overnight maintenance and ground handling constraints that compress available turnaround time.

Weather variables occupy the middle tier of importance. While their linear correlations with delay are weak (Figure 1), the Random Forest is able to capture non-linear threshold effects. For example, delays that spike only when wind speed exceeds a certain level, which explains why wind speed and temperature rank above strictly temporal features such as day of week and month. Precipitation's low importance score (0.030) is consistent with its near-zero linear correlation and further supports the conclusion that ZRH's infrastructure effectively absorbs moderate rainfall events.

4 Conclusion

4.1 Summary of Findings

This project delivered a complete end-to-end data science pipeline for flight delay analysis and prediction at Zurich Airport. The dataset of 29,896 flights across 445 routes and 209 airlines revealed that most departures are on time or only marginally delayed, with a mean delay of 5.44 minutes. A small but operationally significant tail of 12.2% of flights experience delays exceeding 15 minutes, consistent with the reactionary delay patterns typical of a major hub airport.

RQ1: How accurately can departure delays be predicted using operational flight information at Zurich Airport?

Across all four models evaluated, prediction accuracy is real but limited. The best-performing model, Linear Regression trained with weather features, achieves a Mean Absolute Error of 7.07 minutes, an RMSE of 9.54 minutes, and an R^2 of 0.066 on the held-out test set of 5,980 flights. This means the model's predictions are on average 7 minutes away from the true delay, and only 6.6% of the total variance in departure delay can be explained by the available pre-departure features. The remaining 93.4% is driven by factors that are structurally invisible before departure, inbound aircraft arriving late, crew positioning constraints, gate conflicts, and network-level reactionary cascades that no pre-departure dataset can capture. Prediction is therefore possible, but the results confirm a fundamental ceiling on accuracy given the information available at departure time.

RQ2: Does the inclusion of route-level historical delay benchmarks and real-time environmental data improve prediction accuracy compared to a baseline model?

Adding weather variables, temperature, precipitation, and wind speed, to the baseline operational feature set produces a measurable but modest improvement for Linear Regression: R^2 increases from 0.053 to 0.066 (+0.013 absolute, +23.7% relative), MAE falls from 7.12 to 7.07 minutes, and RMSE drops from 9.61 to 9.54 minutes. The improvement for Random Forest is larger in relative terms (R^2 rises from -0.078 to +0.027), though RF still underperforms Linear Regression in all metrics. Route-level historical delay (route_avg_delay), already present in the baseline, is by far the most impactful feature at 41.8% of the Random Forest's importance, confirming that knowing a route's past behaviour is more informative than any weather variable. Weather contributes real but secondary predictive value, consistent with ZRH's operational resilience and the predominantly mild conditions observed during the October–December 2025 period.

RQ3: Which operational and meteorological variables most strongly influence departure delays?

Feature importance extracted from the Random Forest Regressor trained on the full weather-augmented feature set reveals a clear hierarchy. Route average historical delay dominates at 41.8%, reflecting the systematic tendency of structurally delay-prone routes to repeat that pattern. Among weather variables, wind speed (18.6%) and temperature (16.4%) rank second and third, substantially higher than their weak linear correlations ($r = 0.13$ and $r = 0.05$ respectively) would suggest, because the Random Forest captures non-linear threshold effects invisible to correlation analysis. Departure hour accounts for 10.7% of importance, with the peak delay window occurring at 10:00–11:00 UTC (averaging 8.0 minutes) driven by accumulated rotation constraints from the first morning wave. Day of week contributes 6.0%, with Friday averaging the highest delay (6.7 minutes) and Tuesday the lowest (3.8 minutes). Precipitation (3.0%), month (2.6%), and the

weekend flag (0.9%) add negligible predictive value, the weekend effect is already captured by day of week, and month is largely absorbed by the weather variables that track seasonal conditions.

4.2 Limitations

Several limitations constrain the current findings. The three-month observation window (October to December) captures only the autumn-winter period at ZRH, which may not generalise to spring and summer operating conditions. The most impactful drivers of real-world delay, inbound aircraft status, crew availability, gate conflicts, and network reactionary chains, are not present in the pre-departure data available through the AeroDataBox API, creating a structural ceiling on predictive performance. The Random Forest's tendency to overfit on high-cardinality continuous features such as `route_avg_delay` also warrants caution when interpreting feature importance rankings.

4.3 Future Work

Several directions could meaningfully extend this work. Expanding the time window to a full calendar year would allow seasonal effects to be modelled explicitly and improve the robustness of route-level delay benchmarks. Working on this project also proved to be a significant learning experience. Our primary challenge was identifying the right data source, a risk we mitigated by staying flexible, adjusting our approach as needed, and enriching the data with additional sources.

x-rapidapi-host: aerodatabox.p.rapidapi.com 'x-rapidapi-key:
e95175c514mshd8d8f7741ec78a7p112c5ajsnb07307e8f815'